Northeastern University



Introduction to Analytics – ALY6000 20317

Dr. Dee Chiluiza

Module 1 Assignment: Project 1

January 15, 2025

Dev Bhadreskumar Panchal

panchal.devb@northeastern.edu

ALY6000 Project 1 Report

Introduction and Key Findings

In order to accomplish tasks including object creation, vector manipulation, descriptive statistics, and data visualization, this project required utilizing R programming techniques. Among the main conclusions are:

1. **Descriptive Statistics:** Knowledge of statistical metrics like quantiles, mean, median, and standard deviation provide information about the distribution of data.

2. **Analysis of Data Using R**: R's built-in statistical calculation functions and effective coding techniques make data analysis easier.

Importance of Descriptive Statistics

The features and patterns of data are summed up by descriptive statistics. They are crucial for preliminary data exploration because they highlight trends and abnormalities that direct more in-depth investigation.

Advantages of Data Analysis with R

- R's extensive libraries for statistical and graphical techniques make it an effective tool for data analysis.
- The ability to handle a variety of data kinds with flexibility.
- Support from the community and thorough documentation.

Analysis

Objects and Their Benefits: R objects allow for modular and structured coding by storing data and results. Objects like `price`, `total_sales`, and `revenue` made data handling and calculations easier in this project.

Vectors and Their Benefits : It include their ability to handle data sequences effectively. For instance, `speed1` enabled descriptive statistics and visualization by storing vehicle speeds.

Vector operations, like filtering, are quick and computationally effective.

Matrices and Their Benefits: Matrices help in comparison and display by organizing data in tabular form. Revenue and statistics summaries were efficiently created and shown using the `matrix()` method.

Conclusion/Recommendations

Evidence-Based Recommendations :-

1. **Use R for Statistical Analysis:** R is a great option for data-intensive activities due to its statistical capabilities and specialized libraries.
2. **Use Descriptive Statistics:** These measurements offer a fundamental comprehension of the data that is necessary for well-informed choices.

Learning Outcomes

This project enhanced skills in:

- Writing R scripts for data manipulation and analysis.
- Applying statistical concepts in programming.
- Organizing and presenting data using matrices and tables.

References

1. Chiluiza, D. Vectors and Matrices in R, (2024), https://rpubs.com/Dee_Chiluiza/vectors_matrix
2. R Core Team. (2003). The R Project for Statistical Computing : R . https://www.r-project.org/